Logic and Set Theory
Lecture N°1
1 Lic Data Science

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Logic
Statement (Proposition).
Logical Connectives
Conditional statement
Tautology and Contradiction
Algebra of propositions
Theorems

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statement

We need some elementary knowledge about logic for doing mathematics. The central notion is “proposition”.

Definition

A statement (or proposition) is a sentence which declare something. A **statement** is a sentence with is either **true** (T) or **false** (F), but not both simultaneously.

Example

1. "In 1492 Columbus sailed the ocean blue.” (T)
2. "Napoleon won the battle of Waterloo.” (F)

Formula

More generally, by **a formula** we mean a statement, possibly involving some variables, which is either true or false whenever we assign particular values to each of the variables.
We will use capital letters to designate formulas.
If the truth of a formula depends on the values of, say, x, y and z, we will use notation like P(x,y,z) to denote the formula.

Example

If $P(x, y)$ is $x^2 + y = 12$, then $P(2, 8)$ and $P(3, 3)$ are true, while $P(1, 4)$ and $P(0, 6)$ are false. If $Q(x, y, z)$ is $x + y < z$, then $Q(1, 2, 4)$ is true and $Q(2, 3, 4)$ is false.

Whether a sentence is *true* or *false* usually depends on what we are talking about—exactly the same sentence may be true or false depending on the context; for example,

Example

the formula $x|y$ means "x divides y". That is, $x|y$ if there is some z so that $y = xz$. Now, is it true that $3|2$? It depends: if we are talking about integers, the answer is no; if we are talking about rational numbers, the answer is yes, because $2 = 3(2/3)$. (Of course, if $x \neq 0$ and y are any rational numbers then $x|y$, so that it is not a very useful notion. In normal usage, the appearance of the formula $x|y$ implies that x and y are integers.)

Complicated sentences and formulas are put together from simpler ones, using a small number of **logical operations**. Just a handful of these operations will let us say everything we need to say in mathematics.

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We get compound propositions by connecting (simpler) propositions by using **logical connectives**. This is done by means of words "and", "or", "not", or "if . . . then", known from everyday language.

Connective	Symbol	Name
not P	$\neg P$	Negation
P and Q	$P \wedge Q$	conjunction
P or Q	$P \vee Q$	disjunction
if P then Q	$P \Rightarrow Q$	implication
P if and only if Q	$P \Leftrightarrow Q$	equivalence

Definition (Negation)

If P is a formula, then "$notP$" is another formula, which we write symbolically as $\neg P$ or $\overline{P}$. Of course, $\neg P$ is false if P is true and vice versa

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Example

1. "6 is not a prime number" or "It is not true that 6 is prime" or "$\neg$(6 is prime) " (T)
2. "Bouteflika was not a president." (F)

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p	$(\neg P : \overline{P})$
T	F
F	T

Therefore, the double negation corresponds to an affirmation; in other words, the proposals P et $non(non(P))$ are logically equivalent.

Exercise (1: Worksheet N°:1)

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1. $p \Leftrightarrow \overline{\overline{p}}$

P	$\neg P$	$\neg(\neg P)$
T	F	T
F	T	F

Definition (Conjunction: "and")

Suppose that P and Q are formulas. Then "$PandQ$" is a formula written symbolically as $P \wedge Q$, called **the conjunction** of P and Q. For $P \wedge Q$ to be true both P and Q must be true, otherwise it is false.

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Example

1. $5 = 6$ *and* $7 = 8$. **False**
2. Tamanrasset is in the Sahara and Béjaïa is on the Mediterranean coast. **True**
3. Lewis Hamilton is Spanish and Fernando Alonso is British. **False**

Definition (Disjunction: "or")

If P and Q are formulas, then the formula "$PorQ$" is written symbolically as $P \vee Q$, called **the disjunction** of $PandQ$. It is important to note that this is an inclusive **or**, that is, "either or both". So if P, Q or both P and Q are true, so is $P \vee Q$. The only way $P \vee Q$ can be false is if both P and Q are false.

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Example

1. Washington is in Canada or London is in England. (**T**)
2. $5 < 7$ *or* $8 < 10$. (**T**)
3. "Hitler was Spanish or Ghandi was Italian." (**F**)

Exercise (1: Worksheet N°:1)

3) $p \vee q \Leftrightarrow q \vee p$

P	Q	$P \vee Q$	$Q \vee P$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	F

Exercise (1: Worksheet N°:1)

5) $\overline{p \vee q} \Leftrightarrow \bar{p} \wedge \bar{q}$

P	Q	$\neg P$	$\neg Q$	$P \vee Q$	$\neg P \wedge \neg Q$	$\neg(P \vee Q)$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

The next method of combining mathematical statements is slightly more subtle than the preceding ones.

Definition (Implication: $\Rightarrow$)

A statement of the form P implies Q (in symbol $P \Rightarrow Q$) is called an implication. The statement $P \Rightarrow Q$ and the statement 'If P, then Q' are logically same, for (as we shall see) the truth values of both the statements are always same. Again, "P" is called the antecedent or the hypothesis, and "Q" is called the consequent or the conclusion.

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P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

The conditional $P \Rightarrow Q$ is true unless P is true and Q is false.
In other words, property $P \Rightarrow Q$ states that a true proposition cannot imply a false proposition.

Definition ("if and only if": $\Leftrightarrow$)

A statement of the form P if and only if Q (briefly P iff Q) is called an equivalence. P implies and implied by Q (in symbol $P \Leftrightarrow Q$) is logically same as P if and only if Q. We also express it by saying that P is a necessary and sufficient condition for Q.

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p	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Exercise (3: Worksheet N°1)

Complete the dotted lines with the appropriate logical connector: $\Leftrightarrow, \Rightarrow, \Leftarrow$.

1. $x \in \mathbb{R}, x^2 = 4 \cdots x = 2,$
2. $z \in \mathbb{C}, z = \bar{z} \cdots z \in \mathbb{R},$
3. $x \in \mathbb{R}, x = \pi \cdots \exp(2ix) = 1.$

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Example

let P, Q and R be three propositions. We wish to make a truth table for the following statement,

$$(P \Rightarrow R) \wedge (Q \vee \neg R)$$

We can form the true table for this statement, using simple steps, as follows

P	Q	R	$\neg R$	$P \Rightarrow R$	$Q \vee \neg R$	$(P \Rightarrow R) \wedge (Q \vee \neg R)$
T	T	T				

P	Q	R	$\neg R$	$P \Rightarrow R$	$Q \vee \neg R$	$(P \Rightarrow R) \wedge (Q \vee \neg R)$
T	T	T	F			

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T	T	T	F	T		

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T	T	T	F	T	T	

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T	T	T	F	T	T	T
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T	T	T	F	T	T	T
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T	T	T	F	T	T	T
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T	T	T	F	T	T	T
T	T	F	T	F	T	F
T	F	T	F	T	F	F
T	F	F	T	F	T	F
F	T	T	F	T	T	T
F	T	F	T	T	T	T
F	F	T	F	T	F	F
F	F	F	T	T	T	T

We can derive the truth value of proposition $(P \Rightarrow Q) \Leftrightarrow (\neg P \vee Q)$ by means of a truth table:

P	Q	$\neg P$	$(\neg P \vee Q)$	$(P \Rightarrow Q)$	$(P \Rightarrow Q) \Leftrightarrow (\neg P \vee Q)$
T	T	F	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

That is, proposition $(P \Rightarrow Q) \Leftrightarrow (\neg P \vee Q)$ is always true independently from the truth values for P and Q.

It is a so called **tautology**.

We conclude this section with a brief mention of two important concepts.

Definition

A **tautology** is a statement that is true in every valuation of its propositional variables, independent of the truth values assigned to these variables.
In other words, a tautology will contain only T in the last column of its truth table.

Example

The proverbial tautology is $P \vee \neg P$, draw the table truth.

Definition

The negation of a tautology is a **contradiction**, a statement that is necessarily false regardless of the truth values of its propositional variables.

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Example

The statement $P \vee \neg P$ is a contradiction. Give its truth table

Definition

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Remarque

Let P and Q be propositions. Then

- $(P \equiv Q)$ if and only if $(P \Leftrightarrow Q)$ is a tautology,
- P is a contradiction if and only if $\neg P$ is a tautology.

Definition

1. The proposition $Q \Rightarrow P$ is called **the converse** of $P \Rightarrow Q$.
2. The proposition $\neg P \Rightarrow \neg Q$ is called **the inverse** of $P \Rightarrow Q$.
3. The proposition $\neg Q \Rightarrow \neg P$ is called **the contrapositive** of $P \Rightarrow Q$.

Idempotent laws

Let P be a proposition. Then

- $\neg(\neg P) \equiv P$.
- $P \vee P \equiv P, \qquad P \wedge P \equiv P$.
- $P \vee Q \equiv Q \vee P, \qquad P \wedge Q \equiv Q \wedge P,$
- $P \Rightarrow Q \equiv \neg P \vee Q$.

Associative laws Let P, Q and R be propositions. Then

- $(P \vee Q) \vee R \equiv P \vee (Q \vee R),$

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Mathematics consists of propositions of the form: P implies Q, but you never ask whether P is true.

A mathematical statement (theorem, proposition, lemma, corollary) is a proposition of the form $P \Rightarrow Q$. P is called a **sufficient** condition for Q.

A *sufficient* condition P guarantees that proposition Q is true. However, Q can be true even if P is false.

Q is called a **necessary** condition for P, $Q \Leftarrow P$.

A *necessary* condition Q must be true to allow P to be true. It does not guarantee that P is true.

Necessary conditions often are used to find candidates for valid answers to our problems.